

- 9 A car of mass m is travelling with a velocity v .

The velocity of the car is increased by Δv due to a constant resultant force F acting for a time t .

What is a correct expression for t ?

A $t = \frac{mv + m\Delta v}{F}$

B $t = \frac{m\Delta v}{F}$

C $t = \frac{F}{m\Delta v}$

D $t = \frac{F}{mv + m\Delta v}$